$$
\begin{aligned}
&C_{\phi}^{(21)\;(21)\;(12)} \to \\
&\frac{1}{16\pi^2}\Big(m_1\, g_1^6\, \tilde{\mu}^3\, \mathrm{LF}_{3,3,-1}[m_1, \tilde{\mu}] - \frac{2}{3} m_2\, \tilde{\mu}\, g_2^6\, \mathrm{LF}_{2,2,0}[m_2, \tilde{\mu}] + \frac{1}{3} m_2\, \tilde{\mu}\, g_2^6\, \mathrm{LF}_{3,2,-1}[m_2, \tilde{\mu}] - \\
&2 m_2\, \tilde{\mu}\, g_2^6\, \mathrm{LF}_{3,3,-2}[m_2, \tilde{\mu}] + m_2\, \tilde{\mu}\, g_2^6 \left(2 m_2^2 + 3 \tilde{\mu}^2\right) \mathrm{LF}_{3,3,-1}[m_2, \tilde{\mu}] - \\
&\tilde{\mu}\, g_2^4\, y_e^{pr}\, \overline{a_e}^{pr}\, \mathrm{LF}_{4,1,-1}\left[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}\right] + \frac{5}{6} \tilde{\mu}\, g_2^4\, y_e^{pr}\, \overline{a_e}^{pr}\, \mathrm{LF}_{5,1,-2}\left[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}\right] - \\
&3 \tilde{\mu}\, g_2^4\, y_d^{pr}\, \overline{a_d}^{pr}\, \mathrm{LF}_{4,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}\right] + \frac{5}{2} \tilde{\mu}\, g_2^4\, y_d^{pr}\, \overline{a_d}^{pr}\, \mathrm{LF}_{5,1,-2}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}\right] - \\
&\frac{1}{2} \tilde{\mu}\, g_2^4\, y_u^{pr}\, \overline{a_u}^{pr}\, \mathrm{LF}_{2,2,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] - \frac{3}{4} \tilde{\mu}\, g_2^4\, y_u^{pr}\, \overline{a_u}^{pr}\, \mathrm{LF}_{3,1,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] + \\
&\frac{1}{4} \tilde{\mu}\, g_2^4\, y_u^{pr}\, \overline{a_u}^{pr}\, \mathrm{LF}_{3,2,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] + \frac{5}{4} \tilde{\mu}\, g_2^4\, y_u^{pr}\, \overline{a_u}^{pr}\, \mathrm{LF}_{3,1,0}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}\right] + \\
&\frac{1}{2} \tilde{\mu}\, g_2^4\, y_u^{pr}\, \overline{a_u}^{pr}\, \mathrm{LF}_{3,2,-1}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}\right] - \frac{15}{4} \tilde{\mu}\, g_2^4\, y_u^{pr}\, \overline{a_u}^{pr}\, \mathrm{LF}_{4,1,-1}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}\right] + \\
&\frac{5}{2} \tilde{\mu}\, g_2^4\, y_u^{pr}\, \overline{a_u}^{pr}\, \mathrm{LF}_{5,1,-2}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}\right] + \frac{1}{4} m_1\, \tilde{\mu}\, g_1^2\, g_2^4\, \mathrm{LF}_{3,1,0}[\tilde{\mu}, m_1] + \\
&\frac{3}{2} m_1\, \tilde{\mu}\, g_1^4\, g_2^2\, \mathrm{LF}_{3,2,-1}[\tilde{\mu}, m_1] - m_1\, \tilde{\mu}\, g_1^2\, g_2^4\, \mathrm{LF}_{4,1,-1}[\tilde{\mu}, m_1] - \\
&m_1\, \tilde{\mu}\, g_1^4\, g_2^2\, \mathrm{LF}_{4,2,-2}[\tilde{\mu}, m_1] + \frac{5}{6} m_1\, \tilde{\mu}\, g_1^2\, g_2^4\, \mathrm{LF}_{5,1,-2}[\tilde{\mu}, m_1] + \frac{17}{12} m_2\, \tilde{\mu}\, g_2^6\, \mathrm{LF}_{3,1,0}[\tilde{\mu}, m_2] + \\
&\frac{31}{6} m_2\, \tilde{\mu}\, g_2^6\, \mathrm{LF}_{3,2,-1}[\tilde{\mu}, m_2] - 4 m_2\, \tilde{\mu}\, g_2^6\, \mathrm{LF}_{4,1,-1}[\tilde{\mu}, m_2] - 4 m_2\, \tilde{\mu}\, g_2^6\, \mathrm{LF}_{4,2,-2}[\tilde{\mu}, m_2] + \\
&\frac{5}{2} m_2\, \tilde{\mu}\, g_2^6\, \mathrm{LF}_{5,1,-2}[\tilde{\mu}, m_2] + 3 \tilde{\mu}\, g_1^2\, g_2^4\, (m_1 + m_2)\, \mathrm{LF}_{2,2,1,-1}[m_2, \tilde{\mu}, m_1] - \\
&2 \tilde{\mu}\, g_1^2\, g_2^4\, (m_1 + m_2)\, \mathrm{LF}_{3,2,1,-2}[m_2, \tilde{\mu}, m_1] + \tilde{\mu}\, g_2^2\, \overline{y_e}^{st}\, y_e^{pt}\, y_e^{sr}\, \overline{a_e}^{pr}\, \mathrm{LF}_{2,1,1,0}\left[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{s}\right] - \\
&\frac{1}{2} \tilde{\mu}\, g_2^2\, \overline{y_e}^{st}\, y_e^{pt}\, y_e^{sr}\, \overline{a_e}^{pr}\, \mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{s}\right] + 3 \tilde{\mu}\, g_2^2\, \overline{y_d}^{st}\, y_d^{pt}\, y_d^{sr}\, \overline{a_d}^{pr} \\
&\quad \mathrm{LF}_{2,1,1,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}\right] - \frac{3}{2} \tilde{\mu}\, g_2^2\, \overline{y_d}^{st}\, y_d^{pt}\, y_d^{sr}\, \overline{a_d}^{pr}\, \mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}\right] + \\
&3 \tilde{\mu}\, g_2^2\, y_d^{sr}\, \overline{y_u}^{st}\, y_u^{pt}\, \overline{a_d}^{pr}\, \mathrm{LF}_{2,1,1,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{u}}^{t}\right] - \frac{3}{2} \tilde{\mu}\, g_2^2\, y_d^{sr}\, \overline{y_u}^{st}\, y_u^{pt}\, \overline{a_d}^{pr} \\
&\quad \mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{u}}^{t}\right] + \frac{3}{2} \tilde{\mu}\, g_2^2\, \overline{y_d}^{st}\, y_d^{pt}\, y_u^{sr}\, \overline{a_u}^{pr}\, \mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{t}, m_{\tilde{u}}^{r}\right] - \\
&\frac{3}{4} \tilde{\mu}\, g_2^2\, y_u^{sr}\, \overline{a_u}^{pr} \left(\overline{y_d}^{st}\, y_d^{pt} - 3 \overline{y_u}^{st}\, y_u^{pt}\right) \mathrm{LF}_{2,1,1,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}\right] + \\
&\frac{3}{4} \tilde{\mu}\, g_2^2\, y_u^{sr}\, \overline{a_u}^{pr} \left(\overline{y_d}^{st}\, y_d^{pt} - \overline{y_u}^{st}\, y_u^{pt}\right) \mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}\right] + \\
&\frac{3}{2} \tilde{\mu}\, g_2^2\, \overline{y_d}^{st}\, y_d^{pt}\, y_u^{sr}\, \overline{a_u}^{pr}\, \mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, m_{\tilde{d}}^{t}\right] + \\
&\frac{3}{4} \tilde{\mu}\, g_2^2\, \overline{y_d}^{st}\, y_d^{pt}\, y_u^{sr}\, \overline{a_u}^{pr}\, \mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, m_{\tilde{q}}^{s}\right] - \frac{3}{2} \tilde{\mu}\, g_2^2\, y_d^{sr}\, \overline{y_u}^{st}\, y_u^{pt}\, \overline{a_d}^{pr} \\
&\quad \mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}, m_{\tilde{d}}^{r}\right] + \frac{3}{2} \tilde{\mu}\, g_2^2\, y_d^{sr}\, \overline{y_u}^{st}\, y_u^{pt}\, \overline{a_d}^{pr}\, \mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{q}}^{s}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}\right] + \\
&\frac{3}{4} \tilde{\mu}\, g_2^2\, y_u^{sr}\, \overline{a_u}^{pr} \left(\overline{y_d}^{st}\, y_d^{pt} + \overline{y_u}^{st}\, y_u^{pt}\right) \mathrm{LF}_{2,1,1,0}\left[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] + \\
&\frac{3}{4} \tilde{\mu}\, g_2^2\, y_u^{sr}\, \overline{a_u}^{pr} \left(\overline{y_d}^{st}\, y_d^{pt} - \overline{y_u}^{st}\, y_u^{pt}\right) \mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] - \\
&\frac{3}{4} \tilde{\mu}\, g_2^2\, \overline{y_u}^{st}\, y_u^{pt}\, y_u^{sr}\, \overline{a_u}^{pr}\, \mathrm{LF}_{2,2,1,-1}\left[m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}\right] + \\
&\frac{3}{4} \tilde{\mu}\, g_2^2\, y_u^{sr}\, \overline{a_u}^{pr} \left(-\overline{y_d}^{st}\, y_d^{pt} + \overline{y_u}^{st}\, y_u^{pt}\right) \mathrm{LF}_{2,1,1,0}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}\right] + \\
&\frac{3}{4} \tilde{\mu}\, g_2^2\, y_u^{sr}\, \overline{a_u}^{pr} \left(\overline{y_d}^{st}\, y_d^{pt} - \overline{y_u}^{st}\, y_u^{pt}\right) \mathrm{LF}_{3,1,1,-1}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}\right] + \\
&3 m_2\, \tilde{\mu}\, g_1^2\, g_2^4\, \mathrm{LF}_{3,1,1,-1}[\tilde{\mu}, m_1, m_2] + \tilde{\mu}\, g_1^4\, g_2^2\, (2 m_1 + m_2)\, \mathrm{LF}_{3,2,1,-2}[\tilde{\mu}, m_1, m_2] + \\
&\tilde{\mu}\, g_1^4\, g_2^2 \left(m_2\, m_1^2 + \tilde{\mu}^2 (-2 m_1 + m_2)\right) \mathrm{LF}_{3,2,1,-1}[\tilde{\mu}, m_1, m_2] - \\
&3 m_2\, \tilde{\mu}\, g_1^2\, g_2^4\, \mathrm{LF}_{4,1,1,-2}[\tilde{\mu}, m_1, m_2] - \tilde{\mu}\, g_1^2\, g_2^4\, (2 m_1 + m_2)\, \mathrm{LF}_{3,2,1,-2}[\tilde{\mu}, m_2, m_1] + \\
&\tilde{\mu}\, g_1^2\, g_2^4 \left(-2 m_2\, \tilde{\mu}^2 + m_1 \left(3 m_2^2 + \tilde{\mu}^2\right)\right) \mathrm{LF}_{3,2,1,-1}[\tilde{\mu}, m_2, m_1] - \\
&3 \tilde{\mu}\, y_d^{sr}\, \overline{y_u}^{st}\, \overline{y_u}^{uv}\, y_u^{pv}\, y_u^{ut}\, \overline{a_d}^{pr}\, \mathrm{LF}_{1,1,1,1,0}\left[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{u}, m_{\tilde{u}}^{t}\right] - \\
&3 \tilde{\mu}\, \overline{y_d}^{st}\, y_d^{sr}\, y_d^{ut}\, \overline{y_u}^{uv}\, y_u^{pv}\, \overline{a_d}^{pr}\, \mathrm{LF}_{1,1,1,1,0}\left[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}, m_{\tilde{q}}^{u}, m_{\tilde{q}}^{p}\right] - \\
&3 \tilde{\mu}\, \overline{y_d}^{st}\, y_d^{pt}\, y_d^{ur}\, \overline{y_u}^{uv}\, y_u^{sv}\, \overline{a_d}^{pr}\, \mathrm{LF}_{1,1,1,1,0}\left[m_{\tilde{d}}^{t}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{v}\right] - \\
&3 \tilde{\mu}\, \overline{y_d}^{st}\, y_d^{ut}\, \overline{y_u}^{uv}\, y_u^{pv}\, y_u^{sr}\, \overline{a_u}^{pr}\, \mathrm{LF}_{1,1,1,1,0}\left[m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{v}, m_{\tilde{u}}^{r}\right] - \\
&3 \tilde{\mu}\, \overline{y_d}^{st}\, \overline{y_d}^{uv}\, y_d^{pt}\, y_d^{sv}\, y_u^{ur}\, \overline{a_u}^{pr}\, \mathrm{LF}_{1,1,1,1,0}\left[m_{\tilde{d}}^{v}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}\right] + \\
&\tilde{\mu}\, g_2^2\, y_e^{sr}\, \overline{a_e}^{pr}\, \overline{a_e}^{st}\, a_e^{pt}\, \mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{t}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{s}\right] - \\
&\frac{1}{2} \tilde{\mu}\, g_2^2\, y_e^{sr}\, \overline{a_e}^{pr}\, \overline{a_e}^{st}\, a_e^{pt}\, \mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{t}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{s}\right] - \\
&\frac{1}{2} g_2^2\, \tilde{\mu}^3\, \overline{y_e}^{st}\, y_e^{pt}\, y_e^{sr}\, \overline{a_e}^{pr}\, \mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{l}}^{s}, m_{\tilde{e}}^{t}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}\right] + \\
&3 \tilde{\mu}\, g_2^2\, y_u^{pt}\, \overline{a_d}^{pr} \left(a_d^{sr}\, \overline{a_u}^{st} - \tilde{\mu}^2\, y_d^{sr}\, \overline{y_u}^{st}\right) \mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}\right] + \\
&\frac{3}{2} \tilde{\mu}\, g_2^2\, y_u^{pt}\, \overline{a_d}^{pr} \left(-a_d^{sr}\, \overline{a_u}^{st} + \tilde{\mu}^2\, y_d^{sr}\, \overline{y_u}^{st}\right) \mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}\right] + \\
&3 \tilde{\mu}\, g_2^2\, y_d^{sr}\, \overline{a_d}^{pr}\, \overline{a_d}^{st}\, a_d^{pt}\, \mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{t}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}\right] - \\
&\frac{3}{2} \tilde{\mu}\, g_2^2\, y_d^{sr}\, \overline{a_d}^{pr}\, \overline{a_d}^{st}\, a_d^{pt}\, \mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{t}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}\right] - \\
&\frac{3}{2} g_2^2\, \tilde{\mu}^3\, \overline{y_d}^{st}\, y_d^{pt}\, y_u^{sr}\, \overline{a_u}^{pr}\, \mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}\right] - \\
&\frac{3}{2} g_2^2\, \tilde{\mu}^3\, \overline{y_d}^{st}\, y_d^{pt}\, y_u^{sr}\, \overline{a_u}^{pr}\, \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{d}}^{t}, m_{\tilde{u}}^{r}\right] - \\
&3 \tilde{\mu}\, \overline{y_d}^{st}\, y_d^{pt}\, \overline{y_u}^{uv}\, y_u^{sv}\, y_u^{ur}\, \overline{a_u}^{pr}\, \mathrm{LF}_{1,1,1,1,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{q}}^{u}, m_{\tilde{u}}^{r}\right] + \\
&\frac{3}{4} \tilde{\mu}\, g_2^2\, y_u^{sr}\, \overline{a_u}^{pr} \left(3 \overline{a_u}^{st}\, a_u^{pt} + \tilde{\mu}^2\, \overline{y_u}^{st}\, y_u^{pt}\right) \mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}, m_{\tilde{u}}^{r}\right] - \\
&\frac{3}{4} \tilde{\mu}\, g_2^2\, y_u^{sr}\, \overline{a_u}^{pr} \left(\overline{a_u}^{st}\, a_u^{pt} + \tilde{\mu}^2\, \overline{y_u}^{st}\, y_u^{pt}\right) \mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}, m_{\tilde{u}}^{r}\right] - \\
&\frac{3}{2} g_2^2\, \tilde{\mu}^3\, \overline{y_d}^{st}\, y_d^{pt}\, y_u^{sr}\, \overline{a_u}^{pr}\, \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{s}\right] - \\
&\frac{3}{4} g_2^2\, \tilde{\mu}^3\, \overline{y_u}^{st}\, y_u^{pt}\, y_u^{sr}\, \overline{a_u}^{pr}\, \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}\right] + \\
&\frac{3}{2} \tilde{\mu}\, g_2^2\, y_u^{pt}\, \overline{a_d}^{pr} \left(-a_d^{sr}\, \overline{a_u}^{st} + \tilde{\mu}^2\, y_d^{sr}\, \overline{y_u}^{st}\right) \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}\right] - \\
&\frac{3}{4} g_2^2\, \tilde{\mu}^3\, \overline{y_u}^{st}\, y_u^{pt}\, y_u^{sr}\, \overline{a_u}^{pr}\, \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}\right] + \\
&\frac{3}{2} \tilde{\mu}\, g_2^2\, y_d^{sr}\, \overline{a_d}^{pr}\, \overline{a_u}^{st}\, a_u^{pt}\, \mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{q}}^{s}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}\right] - \\
&\frac{3}{2} g_2^2\, \tilde{\mu}^3\, \overline{y_d}^{st}\, y_d^{pt}\, y_d^{sr}\, \overline{a_d}^{pr}\, \mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{q}}^{s}, m_{\tilde{d}}^{t}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}\right] - \\
&\frac{3}{2} \tilde{\mu}\, g_2^2\, y_u^{pt}\, \overline{a_d}^{pr}\, a_d^{sr}\, \overline{a_u}^{st}\, \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{u}}^{t}\right] + \\
&\frac{3}{4} \tilde{\mu}\, g_2^2\, y_u^{sr}\, \overline{a_u}^{pr} \left(\overline{a_u}^{st}\, a_u^{pt} - \tilde{\mu}^2\, \overline{y_u}^{st}\, y_u^{pt}\right) \mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}, m_{\tilde{u}}^{r}\right] - \\
&\frac{3}{4} \tilde{\mu}\, g_2^2\, y_u^{sr}\, \overline{a_u}^{pr} \left(\overline{a_u}^{st}\, a_u^{pt} + \tilde{\mu}^2\, \overline{y_u}^{st}\, y_u^{pt}\right) \mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}, m_{\tilde{u}}^{r}\right] - \\
&\frac{3}{4} \tilde{\mu}\, g_2^2\, y_u^{sr}\, \overline{a_u}^{pr}\, \overline{a_u}^{st}\, a_u^{pt}\, \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}\right] + \\
&\frac{3}{2} \tilde{\mu}\, g_2^2\, y_d^{sr}\, \overline{a_d}^{pr}\, \overline{a_u}^{st}\, a_u^{pt}\, \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}\right] - \\
&\frac{3}{4} \tilde{\mu}\, g_2^2\, y_u^{sr}\, \overline{a_u}^{pr}\, \overline{a_u}^{st}\, a_u^{pt}\, \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}\right] + \\
&\frac{3}{4} \tilde{\mu}\, g_2^2\, y_u^{sr}\, \overline{a_u}^{pr} \left(\overline{a_u}^{st}\, a_u^{pt} + \tilde{\mu}^2\, \overline{y_u}^{st}\, y_u^{pt}\right) \mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}\right] - \\
&\frac{3}{4} \tilde{\mu}\, g_2^2\, y_u^{sr}\, \overline{a_u}^{pr} \left(\overline{a_u}^{st}\, a_u^{pt} + \tilde{\mu}^2\, \overline{y_u}^{st}\, y_u^{pt}\right) \mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}\right] + \\
&\frac{3}{4} \tilde{\mu}\, g_2^2\, y_u^{sr}\, \overline{a_u}^{pr} \left(\overline{a_u}^{st}\, a_u^{pt} + \tilde{\mu}^2\, \overline{y_u}^{st}\, y_u^{pt}\right) \mathrm{LF}_{2,1,1,1,0}\left[m_{\tilde{u}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}\right] - \\
&\frac{3}{4} \tilde{\mu}\, g_2^2\, y_u^{sr}\, \overline{a_u}^{pr} \left(\overline{a_u}^{st}\, a_u^{pt} + \tilde{\mu}^2\, \overline{y_u}^{st}\, y_u^{pt}\right) \mathrm{LF}_{3,1,1,1,-1}\left[m_{\tilde{u}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}\right] - \\
&\frac{3}{4} \tilde{\mu}\, g_2^2\, y_u^{sr}\, \overline{a_u}^{pr} \left(\overline{a_u}^{st}\, a_u^{pt} + \tilde{\mu}^2\, \overline{y_u}^{st}\, y_u^{pt}\right) \mathrm{LF}_{2,2,1,1,-1}\left[m_{\tilde{u}}^{t}, m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}\right] - \\
&3 \tilde{\mu}\, y_d^{sr}\, y_d^{ut}\, \overline{y_u}^{uv}\, \overline{a_d}^{pr}\, \overline{a_d}^{st}\, a_u^{pv}\, \mathrm{LF}_{1,1,1,1,1,0}\left[m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{v}\right] - \\
&3 \tilde{\mu}\, y_d^{pt}\, \overline{y_u}^{uv}\, y_u^{sv}\, \overline{a_d}^{pr}\, \overline{a_d}^{st}\, a_d^{ur}\, \mathrm{LF}_{1,1,1,1,1,0}\left[m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{u}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}\right] - \\
&3 \tilde{\mu}\, \overline{y_d}^{uv}\, y_d^{pv}\, y_u^{ut}\, \overline{a_d}^{pr}\, a_d^{sr}\, \overline{a_u}^{st}\, \mathrm{LF}_{1,1,1,1,1,0}\left[m_{\tilde{d}}^{r}, m_{\tilde{d}}^{v}, m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}\right] - \\
&3 \tilde{\mu}\, y_d^{ur}\, \overline{y_u}^{uv}\, y_u^{sv}\, \overline{a_d}^{pr}\, \overline{a_u}^{st}\, a_u^{pt}\, \mathrm{LF}_{1,1,1,1,1,0}\left[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}, m_{\tilde{u}}^{v}\right] - \\
&3 \tilde{\mu}\, \overline{y_d}^{uv}\, y_d^{sv}\, y_d^{ur}\, \overline{a_d}^{pr}\, \overline{a_u}^{st}\, a_u^{pt}\, \mathrm{LF}_{1,1,1,1,1,0}\left[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{u}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}\right] + \\
&3 \tilde{\mu}\, \overline{y_u}^{uv}\, y_u^{pv}\, y_u^{ut}\, \overline{a_d}^{pr} \left(-a_d^{sr}\, \overline{a_u}^{st} + \tilde{\mu}^2\, y_d^{sr}\, \overline{y_u}^{st}\right) \mathrm{LF}_{1,1,1,1,1,0}\left[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{u}, m_{\tilde{u}}^{t}\right] + \\
&3 \tilde{\mu}^3\, \overline{y_d}^{st}\, y_d^{pt}\, y_d^{ur}\, \overline{y_u}^{uv}\, y_u^{sv}\, \overline{a_d}^{pr}\, \mathrm{LF}_{1,1,1,1,1,0}\left[m_{\tilde{d}}^{t}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{v}\right] + \\
&3 \tilde{\mu}^3\, \overline{y_d}^{st}\, y_d^{pt}\, y_d^{ur}\, \overline{y_u}^{uv}\, y_u^{sv}\, \overline{a_d}^{pr}\, \mathrm{LF}_{1,1,1,1,1,0}\left[m_{\tilde{d}}^{t}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{u}, m_{\tilde{u}}^{v}\right] + \\
&3 \tilde{\mu}^3\, \overline{y_d}^{st}\, y_d^{ut}\, \overline{y_u}^{uv}\, y_u^{pv}\, y_u^{sr}\, \overline{a_u}^{pr}\, \mathrm{LF}_{1,1,1,1,1,0}\left[m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{v}, m_{\tilde{u}}^{r}\right] + \\
&3 \tilde{\mu}^3\, \overline{y_d}^{st}\, y_d^{ut}\, \overline{y_u}^{uv}\, y_u^{pv}\, y_u^{sr}\, \overline{a_u}^{pr}\, \mathrm{LF}_{1,1,1,1,1,0}\left[m_{\tilde{d}}^{t}, m_{\tilde{q}}^{u}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{v}, m_{\tilde{u}}^{r}\right] + \\
&3 \tilde{\mu}^3\, \overline{y_d}^{st}\, \overline{y_d}^{uv}\, y_d^{pv}\, y_d^{sr}\, y_d^{ut}\, \overline{a_d}^{pr}\, \mathrm{LF}_{1,1,1,1,1,0}\left[m_{\tilde{d}}^{v}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{q}}^{u}\right] + \\
&3 \tilde{\mu}^3\, \overline{y_d}^{st}\, \overline{y_d}^{uv}\, y_d^{pt}\, y_d^{sv}\, y_u^{ur}\, \overline{a_u}^{pr}\, \mathrm{LF}_{1,1,1,1,1,0}\left[m_{\tilde{d}}^{v}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{q}}^{u}, m_{\tilde{u}}^{r}\right] - \\
&3 \tilde{\mu}\, \overline{y_d}^{uv}\, y_u^{sr}\, y_u^{ut}\, a_d^{pv}\, \overline{a_u}^{pr}\, \overline{a_u}^{st}\, \mathrm{LF}_{1,1,1,1,1,0}\left[m_{\tilde{d}}^{v}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}\right] + \\
&\tilde{\mu}^3\, \overline{y_e}^{st}\, \overline{y_e}^{uv}\, y_e^{pv}\, y_e^{sr}\, y_e^{ut}\, \overline{a_e}^{pr}\, \mathrm{LF}_{1,1,1,1,1,0}\left[m_{\tilde{e}}^{v}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}, m_{\tilde{l}}^{s}, m_{\tilde{l}}^{u}\right] + \\
&3 \tilde{\mu}^3\, \overline{y_u}^{st}\, \overline{y_u}^{uv}\, y_u^{pv}\, y_u^{sr}\, y_u^{ut}\, \overline{a_u}^{pr}\, \mathrm{LF}_{1,1,1,1,1,0}\left[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{q}}^{u}, m_{\tilde{u}}^{v}, m_{\tilde{u}}^{r}\right] - \\
&3 \tilde{\mu}\, \overline{y_d}^{uv}\, y_d^{pv}\, y_u^{sr}\, \overline{a_u}^{pr}\, \overline{a_u}^{st}\, a_u^{ut}\, \mathrm{LF}_{1,1,1,1,1,0}\left[m_{\tilde{q}}^{u}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}, m_{\tilde{u}}^{r}\right] + \\
&3 \tilde{\mu}^3\, \overline{y_d}^{uv}\, y_d^{pt}\, y_d^{sv}\, \overline{a_d}^{pr}\, \overline{a_d}^{st}\, a_d^{ur}\, \mathrm{LF}_{1,1,1,1,1,1,0}\left[m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{d}}^{v}, m_{\tilde{q}}^{u}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}\right] + \\
&3 \tilde{\mu}^3\, y_d^{sr}\, y_d^{ut}\, \overline{y_u}^{uv}\, \overline{a_d}^{pr}\, \overline{a_d}^{st}\, a_u^{pv}\, \mathrm{LF}_{1,1,1,1,1,1,0}\left[m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{q}}^{u}, m_{\tilde{u}}^{v}\right] + \\
&3 \tilde{\mu}^3\, \overline{y_d}^{uv}\, y_d^{pv}\, y_u^{ut}\, \overline{a_d}^{pr}\, a_d^{sr}\, \overline{a_u}^{st}\, \mathrm{LF}_{1,1,1,1,1,1,0}\left[m_{\tilde{d}}^{r}, m_{\tilde{d}}^{v}, m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{u}, m_{\tilde{u}}^{t}\right] + \\
&3 \tilde{\mu}^3\, y_d^{ur}\, \overline{y_u}^{uv}\, y_u^{sv}\, \overline{a_d}^{pr}\, \overline{a_u}^{st}\, a_u^{pt}\, \mathrm{LF}_{1,1,1,1,1,1,0}\left[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{u}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}, m_{\tilde{u}}^{v}\right] - \\
&3 \tilde{\mu}\, y_u^{ut}\, \overline{a_d}^{pr}\, a_d^{sr}\, \overline{a_u}^{st}\, \overline{a_u}^{uv}\, a_u^{pv}\, \mathrm{LF}_{1,1,1,1,1,1,0}\left[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{u}, m_{\tilde{u}}^{v}, m_{\tilde{u}}^{t}\right] - \\
&3 \tilde{\mu}^5\, \overline{y_d}^{st}\, y_d^{pt}\, y_d^{ur}\, \overline{y_u}^{uv}\, y_u^{sv}\, \overline{a_d}^{pr}\, \mathrm{LF}_{1,1,1,1,1,1,0}\left[m_{\tilde{d}}^{t}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{u}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{v}\right] - \\
&3 \tilde{\mu}\, y_d^{sr}\, \overline{a_d}^{pr}\, \overline{a_d}^{st}\, a_d^{ut}\, \overline{a_u}^{uv}\, a_u^{pv}\, \mathrm{LF}_{1,1,1,1,1,1,0}\left[m_{\tilde{d}}^{t}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{u}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{v}\right] - \\
&3 \tilde{\mu}^5\, \overline{y_d}^{st}\, y_d^{ut}\, \overline{y_u}^{uv}\, y_u^{pv}\, y_u^{sr}\, \overline{a_u}^{pr}\, \mathrm{LF}_{1,1,1,1,1,1,0}\left[m_{\tilde{d}}^{t}, m_{\tilde{q}}^{u}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{v}, m_{\tilde{u}}^{r}\right] + \\
&3 \tilde{\mu}^3\, \overline{y_d}^{uv}\, y_u^{sr}\, y_u^{ut}\, a_d^{pv}\, \overline{a_u}^{pr}\, \overline{a_u}^{st}\, \mathrm{LF}_{1,1,1,1,1,1,0}\left[m_{\tilde{d}}^{v}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{q}}^{u}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}\right] + \\
&\tilde{\mu}^3\, \overline{y_e}^{uv}\, y_e^{pt}\, y_e^{sv}\, \overline{a_e}^{pr}\, \overline{a_e}^{st}\, a_e^{ur}\, \mathrm{LF}_{1,1,1,1,1,1,0}\left[m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}, m_{\tilde{e}}^{v}, m_{\tilde{l}}^{u}, m_{\tilde{l}}^{p}, m_{\tilde{l}}^{s}\right] + \\
&3 \tilde{\mu}^3\, \overline{y_u}^{uv}\, y_u^{pt}\, y_u^{sv}\, \overline{a_u}^{pr}\, \overline{a_u}^{st}\, a_u^{ur}\, \mathrm{LF}_{1,1,1,1,1,1,0}\left[m_{\tilde{q}}^{u}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}, m_{\tilde{u}}^{v}\right]\Big)
\end{aligned}
$$